

Sr no.	Required Technical Specifications	Qty	Offered Technical Specifications
1	GALVANIZED	2	
	50 m3 Galvanized Hot-Pressed Steel Sectional Tank for storing drinking water, with external flanges & weatherproof cover.		
	Panel material specification grade ISO1461:2022.		
	Panel flange joints: Bolted combined double flanged (45 to 90 degrees) to the face of the panels.		
	Roof panel covers weather and dust proof with a hot-dipped galvanized (ISO1461:2022) cover width of 600 square.		
	lockable manhole and vermin-proof 100-diameter air vents.		
	Internal Reinforcement: all brackets, Angle bracing, bolts and nuts, trusses and supports are of hot-dipped galvanized.		
	internal and external ladders of hot-dipped galvanized type bolted on the manhole cover and side of the wall.		
	level indicator; hot-dipped galvanized type (calibrated in metric and paint-coated).		
	Pipe connections and nozzles: provide an inlet. outlet and overflow connection to the tank.		
	The expected dimension: length 5m x width 5m x height 2m		
	The tank will be on an elevated concrete tower. With a wind resistance of up to 140 km/h is recommended.		
	2 Set/tank HDG flange nozzles + bolt, nuts, rubber gaskets: see details below for Inlet, outlet, drain (100 DIA), and overflow (150 DIA).		
	Manhole: 500 x 500 mm with cover and gasket.		
	Inlet: 100mm flanged or threaded connection.		
	Outlet: 100 mm drain fitting at the lowest point.		
	Overflow: 150 mm.		
	16 Epoxy paint for inside tank for potable water storage 1st coating: HB epoxy + Hardener 5 ltrs per tank.		
	Epoxy coating must be suitable for drinking water. referring to NSF61 or WRAS approved.		
	10 Epoxy paint for inside tank for potable water storage 2nd coating: Etching primer 5 ltrs + Hardener per tank. Epoxy coating must be suitable for drinking water. referring to NSF61 or WRAS approved.		
	2 Outside paint / tank : Nippon Polyurethane Recoat able (4.5 ltr + 0.5 ltr hardener) + EA9 primer as undercoat (4.0 ltr + 1.0 ltr hardener)		
	Additional Components		
	Vent with mosquito-proof mesh (UNICEF WASH standard practice).		
	Internal ladder (FRP or stainless steel).		
	External ladder + safety cage.		
	Liquid level indicator (mechanical float or graduated gauge).		

	Lifting lugs and base support pads (per manufacturer).		
	Anchor bolts for high wind zones.		
	Manufacturer must supply: Engineering drawings with dimensions and laminate schedule.		
	Material data sheets for GHP.		
	Certificates of compliance with standards.		
	Installation instructions.		
	PACKING, MARKING AND TRANSPORT		
	Tank shall be wrapped in protective plastic sheeting to prevent UV and mechanical damage.		
	All nozzles/flanges protected with temporary covers		
	Package marking shall follow UNICEF supply guidelines:		
	Item name		
	Volume (50,000 L)		
	Gross weight		
	Fragile/Handle with Care labeling		
	Provide accessories in a separate, clearly labelled box.		
	Transport must ensure no distortion of the shell.		
2	50 m ³ (50,000 Litre) Fibre Reinforced Polymer (FRP) for Potable Water Storage.	2	
	1. GENERAL REQUIREMENTS		
	1.1 Scope: This specification covers the design, manufacture, testing, and supply of a 50,000 litre Rectangular FRP potable water storage tank suitable for potable water storage in humanitarian WASH programmes. The tank shall be supplied complete with all structural components, fittings, access ladder, manhole, and installation accessories.		
	1.2 Standards and Reference Documents		
	The FRP tank shall comply with the most recent editions of:		
	ASTM D 3299 # Filament-wound GRP chemical-resistant tanks.		
	ASTM C 582 # Contact-molded FRP laminates		
	ASTM D 4097 # Contact-molded GRP tanks		
	AWWA D 120 # Thermosetting FRP tanks		
	2. DESIGN REQUIREMENTS		
	2.1 Tank Configuration		
	Vertical, closed top, rectangular FRP tank		
	Flat base with dome shaped top		
	Nominal capacity: 50 m ³ (50,000 litres).		
	Designed proportionally to achieve the required volume and structural integrity to fit length 5m x width 5m x height 2m		
	Minimum structural laminate thickness: # 3/8 inch (9.5 mm) (excluding corrosion barrier)		
	2.2 Operating Parameters		
	Working pressure: atmospheric (solar pump-fed).		
	Operating temperature: 5#45°C		
	UV stabilized for outdoor installation in tropical climates.		

	Tank roof shall withstand # 50 lb/ft ² (uniform load).		
	2.3 Structural Strength		
	Designed to safely support static and dynamic loading from water and wind.		
	FRP laminate must be corrosion-resistant resin with high tensile and flexural strength.		
	Stress rupture requirement: minimum 1,000 hours at 150 psi and 150°F (material performance)		
	3. MATERIALS AND CONSTRUCTION		
	3.1 Resin System		
	Corrosion-resistant resin throughout corrosion barrier and structural layers.		
	No fire-retardant or antimony compounds in corrosion barrier layers.		
	Halogenated resin permitted for outer flame spread compliant layers when required.		
	3.2 Laminate Structure		
	Corrosion barrier (inner surface in contact with water):		
	Two resin-rich veil layers + three layers chopped strand mat.		
	Total thickness: 20#30 mil (0.5#0.75 mm).		
	Composition: 90% resin / 10% veil (by weight).		
	Structural laminate:		
	Filament wound glass fibres.		
	Thickness per design but # 3/8 inch structural.		
	Must comply with ASME RTP 1 Type II laminate structure.		
	3.3 External Surface UV stabilized gel coat.		
	Colour: natural resin, blue, grey, or specified (manufacturer standard colours).		
	4. FITTINGS AND ACCESSORIES		
	4.1 Openings Manhole:		
	Diameter # 500 mm with cover and gasket.		
	Inlet: 75#100 mm flanged or threaded connection.		
	Outlet: 75#100 mm; drain fitting at lowest point.		
	Overflow: 75 mm minimum.		
	4.2 Additional Components		
	Vent with mosquito-proof mesh (UNICEF WASH standard practice).		
	Internal ladder (FRP or stainless steel).		
	External ladder + safety cage.		
	Liquid level indicator (mechanical float or graduated gauge).		
	Lifting lugs and base support pads (per manufacturer).		
	Anchor bolts for high wind zones.		
	5. QUALITY ASSURANCE AND TESTING		
	5.1 Factory Testing		
	The tank shall undergo: Hydrostatic testing at full capacity before dispatch.		
	Visual inspection according to ASTM D 2563 (defect classification).		

	Verification of laminate hardness (Barcol hardness test # ASTM D 2583).		
	5.2 Documentation		
	Manufacturer must supply: Engineering drawings with dimensions and laminate schedule.		
	Material data sheets for resin and fibre.		
	Certificates of compliance with standards.		
	Installation instructions.		
	6. PACKING, MARKING AND TRANSPORT		
	Tank shall be wrapped in protective plastic sheeting to prevent UV and mechanical damage.		
	All nozzles/flanges protected with temporary covers.		
	Package marking shall follow UNICEF supply guidelines:		
	Item name		
	Volume (50,000 L)		
	Gross weight		
	Fragile/Handle with Care labeling		
	Provide accessories in a separate, clearly labelled box.		
	Transport must ensure no distortion of shell.		
	7. INSTALLATION REQUIREMENTS		
	Install on a level reinforced concrete platform sized per manufacturer guidelines.		
	Anchor bolts must be fixed before tank placement.		
	All joints sealed using NSF approved sealants (for potable water).		
	Follow supplied manufacturer installation manual.		
	8. EXPECTED SERVICE LIFE		
	Minimum 15#20 years under tropical outdoor conditions when properly installed and maintained.		
	FRP is inherently corrosion resistant and requires minimal maintenance.		